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Directional solidification of Al–1.5 wt% Ni alloys under diffusion transport in space and fluid-flow localisation on earth

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Abstract

We present results of a comparative experimental study in Space (diffusion transport) and on Earth (convection and diffusion transport) of solute macrosegregation and directional solidification microstructure in bulk Al–1.5 wt% Ni alloys, which possess a stabilising solutal effect in directional upward solidification on ground. The growth parameters are in the cell–dendrite transition so that cells (no sidebranches) and dendrites (sidebranches) are simultaneously observed, the later having larger spacing. In the benchmark microstructures solidified in microgravity, it is found for the first time that the conditions for the analogy with viscous fingering are satisfied (spacing Peclet number $\ll 1$ and tip constitutional supercooling of order one). The experimental primary spacing is about half the numerical prediction by Lu and Hunt's model. It is suggested that the full panorama of the symmetric and non-symmetric cellular and dendritic branches may explain the misfit. Also, the longitudinal solute macrosegregation across the mushy zone is discussed and modelled, which imposes to depart from the classical Flemings profile. Due to fluid flow driven by the radial temperature gradient, the ground samples show a large macroscopic deformation of the growth front concomitant with microstructure localisation. Cells then appear in the core and dendrites at the periphery. In addition, the measured primary spacing in the centre is smaller than in the corresponding microgravity experiments, which enables to estimate the downward component of the flow velocity in this region.
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1. Introduction

Directional solidification of binary alloys is a pattern-forming system archetype of technological importance in casting. Three parameters control the morphology of the solid–liquid interface during an experiment, namely initial solute concentration C_0, temperature gradient G at the interface in the liquid and growth rate V. For given values of C_0 and G, the solidification front is planar at slow growth velocities. As the rate is increased above a critical value, the smooth interface breaks down into a cellular microstructure, which becomes dendritic at higher growth velocities. Those interface patterns result from the Mullins–Sekerka instability [1] and much effort has been devoted to the prediction of the cell-, or dendrite-, array characteristics as a function of the solidification control parameters [2–6].

However, natural convection in the melt is well known to be the major source of various disturbing effects which can significantly modify or mask important physical mechanisms [7–9]. In particular, fluid flow can be caused by any density gradient in the melt that is not suitably oriented with respect to the gravity vector. The thermo-solutal origin of the front deformation was subsequently proved in different studies in which the orientation of growth relative to gravity was systematically varied: hypo- and hyper-eutectic Al–Cu [10], Pb–Au or Pb–Sn [11,12] and Pb–Tl [13] alloys. Thus, attempts have often been made to eliminate convection in the liquid phase in directional solidification studies. A simple way of doing this is to ensure that the density gradient is vertical everywhere, the heaviest material being at the bottom. This situation is more or less approached when growth takes place upwards in an alloy system in which the rejected (absorbed) solute is denser (lighter) than the solvent. Actually, even in this experimental configuration, flow patterns adjacent to the front are known to develop under the effect of any horizontal density gradient, as revealed by the macroscopic front deformation and lateral microstructure heterogeneity observed at low solidification rates. A first qualitative evidence of these effects was given in Ref. [14] where front shapes were studied as a function of the solidification rate in several Al–Cu alloys solidified above the morphological instability threshold. Since, similar effects have been observed below this threshold both in transparent [15] and semiconductor systems. In addition, the influence of convection on the dendritic microstructure was demonstrated in Ref. [10] by comparing the patterns obtained on Earth with those formed under identical solidification conditions in a microgravity environment. It was in particular found that in such a configuration with a highly permeable mush, the fluid flow was able to strongly reduce the primary dendrite spacing in the regions where it was locally oriented towards the mush.

As stressed for instance in Ref. [12], a comparative study between samples solidified on Earth and in Space, under reduced gravity, is also particularly worth in the range of the cell–dendrite transition, since it generally occurs when the solidification rate is lower than the fluid-flow velocity so that the convective effects are pronounced. This is the reason why microgravity experiments were previously performed on Al–Li alloys in the neighbourhood of the cellular–dendritic transition [16]. For that case where the rejected solute was lighter (i.e. the configuration was solutally destabilising in upward solidification on ground), both a reduction of primary spacing, and a transition towards cellular structures under the effect of convection were evidenced. In the present work, a similar study is performed on Al–1.5 wt% Ni, for which the rejected solute is denser. Another strong difference with the previous study concerns the value of the solute segregation coefficient ($\ll 1$ in the present case), which is one of the material's parameters controlling the distance between the morphological instability threshold and the cell–dendrite transition.

After briefly recalling the experimental procedure, results are presented and discussed successively in terms of solute segregation at the scale of the sample, and microstructure of the solid–liquid interface (both primary spacing and morphology).

2. Experimental procedure

Directional solidification was carried out on Al–1.5 wt% Ni alloys in both terrestrial and

microgravity conditions in the Advanced Gradient Heating Facility (AGHF) of the European Space Agency (ESA). For ground experiments, samples were solidified vertically upwards in a thermal and solutal stabilising configuration with respect to convection driven by the axial gradients, so that fluid flow is induced by the inevitable radial temperature gradient caused by the difference in thermal conductivity between the liquid and solid phases. Microgravity experiments were conducted during the LMS and STS-95 space missions in the American space shuttle Columbia.

The AGHF facility is a Bridgman furnace made of two heaters and a cooling zone which impose a thermal gradient along the axis of the device. These elements can be moved over a fixed cartridge containing the Al–1.5 wt% Ni sample placed inside a boron nitride crucible. The experiments were performed in vacuum. To measure the temperature gradient and solidification rate throughout the process, the cartridges were equipped with 12 thermocouples, glued on the outer surface of the crucible. For the LMS and STS-95 space missions, the alloy specimens were rods of 8 and 10 mm in diameter and the thermal gradients in the liquid were 35 ± 2 and 24 ± 2 K/cm respectively.

The heating-up stage began with the furnace at the lowest position so that about 40 mm of the sample remained solid when the nominal temperature was reached. Controlled solidification started after a thermal stabilisation period of roughly 1 h. In one pair of experiments (FM2/GM2), a single low solidification velocity was imposed whereas in the two other pairs of experiments (FM1/GM1 and FM3/GM3), two growth rates were applied successively. The solidification parameters of all experiments are summarised in Table 1. It is worth noting that sample shrinkage occurred during the LMS mission, due to the absence of hydrostatic pressure. Consequently, for the subsequent STS-95 space mission, a piston device was added in order to ensure that the solidifying specimen kept a constant diameter and remained in full contact with the crucible.

At the end of the solidification period, the melt was quenched by rapid furnace pulling ($V > 150\,\mu m/s$). This velocity was roughly one hundred times higher than the solidification rate so that the interface microstructure was frozen.

After the experiment, X-ray radiography of the sample was performed, which qualitatively reveals the longitudinal segregation thanks to the strong absorption contrast between Ni and Al. The segregation profile was qualitatively determined by recording the grey-level variation of the radiograph along the sample axis, which also allowed in locating the position of the solidification front at the successive stages of the experiment (i.e. end of stabilisation, velocity change if any, and quench). The solute concentration profile was then calibrated both by chemical analysis of thin slices (1 mm thick) cut along the sample, and by EDS analysis. These analyses were performed in the unmelted part, the solidification zone and the quenched liquid.

To obtain the morphology of the quenched solidification front, a metallographic examination was first performed on a longitudinal section. Then, a slice containing the solid–liquid interface was cut and polished to obtain a sequence of closely-spaced transverse sections through the solid–liquid zone. Image analysis techniques [17] were applied on optical micrographs of selected

Table 1
Experimental control parameters of the present study (FM for experiments in the AGHF Flight Model, GM for experiments in the AGHF Ground Model)

Sample (μg)	Thermal gradient (K/cm)	Pulling rate (μm/s)	Sample (1 g)	Thermal gradient (K/cm)	Pulling rate (μm/s)
FM1-1	34	3.83	GM1-1	35	3.83
FM1-2	34	1.83	GM1-2	35	1.79
FM2	34	0.83	GM2	33	1.0
FM3-1	24	2.33	GM3-1	25	2.67
FM3-2	24	1.5	GM3-2	25	1.67

regions in order to determine the histograms of primary spacing.

3. Solute segregation at the macroscopic scale

Nickel segregation at the scale of the sample, and its effects on microstructure, are first examined. The relative influence of convection is derived by comparing the results obtained in microgravity and on ground samples.

3.1. Longitudinal solute macrosegregation

3.1.1. Experimental profile

The average solute concentration measured along the solidification direction by chemical analysis of slices of 1 mm in thickness is shown in Fig. 1a for the microgravity and ground samples FM2 and GM2 ($V = 0.83\,\mu m/s$). In both cases, there is a plateau of almost constant concentration equal to the initial concentration Al–1.5 wt% Ni, measured in the unmelted part of the sample. Actually, as discussed in details elsewhere [18], a complex concentration profile is observed in the region between the unmelted part of the sample and the plateau of constant concentration due to Temperature Gradient Zone Melting (TGZM) migration of Ni-rich liquid channels and inclusions during the preparatory melting and thermal stabilisation stages that preceded directional solidification (see Fig. 1 in Ref. [18]). In this part of the sample stationary growth was thus reached after only about 1 cm of directional growth, i.e. on a time-scale that is 36 times faster than would be expected for the solute recoil of a planar front, $D/(kV^2) = 4.4 \times 10^5$ s with D the solute diffusion coefficient in the liquid ($D = 3 \times 10^{-5}\,cm^2/s$) and k the solute partition coefficient which is very small for dilute Al–Ni alloys, $k = 0.0087$. This should be emphasised because it invalidates the common believing, based on planar front considerations, that a very low value of the solute segregation coefficient precludes the investigation of steady-state cellular/dendritic microstructures on samples of usual length, 10–20 cm. The explanation is likely that the cell/dendrite tips actually advance ahead of the planar front to a position rather close to the liquidus. Consequently, the recoil of the tip envelope should rather be considered, with the large planar-front recoil component being almost compensated by the large tip protrusion in microstructure formation in the present series of Al–Ni experiments. The concentration variation measured near the quenched front indicates the presence of an enriched layer. The overall concentration profile across this region is given by the variation of the optical density along an X-ray radiograph (Fig 1b).

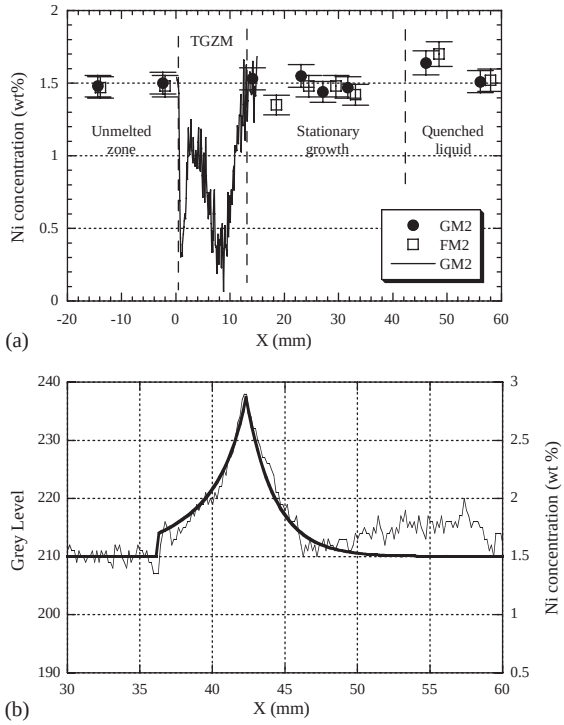

Fig. 1. (a) Profiles of Ni longitudinal macrosegregation in microgravity (FM2) and ground (GM2) samples (chemical analysis of thin slices cut all along the samples), with the variation in the TGZM region superimposed (scan line). (b) Variation of Ni concentration across the quenched mushy zone deduced from the absorption contrast on an X-ray radiograph, and profile predicted by modelling in Section 3.1.2.

3.1.2. Solute segregation accompanying cellular/dendritic growth under diffusion control

The characteristic profile shown on Fig. 1b results from the solute diffusion occurring both in

the liquid ahead the front and in the interdendritic or intercellular liquid. Starting from the left, the profile includes the following five features: (i) a plateau at the bulk concentration C_0 corresponding to the fully solid stationary pattern, (ii) a concentration jump at the position of the eutectic front, (iii) a rising part in the mushy zone, (iv) a maximum at the position of the cell/dendrite tips at the time of the quench, and (v) a falling part corresponding to the boundary layer formed ahead of the tips by the rejected solute. The extent δ of this layer is equal to the classical diffusion length $\delta = D/V$ with D the diffusion coefficient in the liquid.

Step ii: For k approaching zero so that the primary dendrites are almost pure aluminium ($C_S^{Al} = 0$), the global measured solute jump ΔC_g in Fig. 1b is given by

$$\Delta C_g = f_{L,E}\Delta C_E = f_{L,E}(C_L^E - C_S^E) \tag{1a}$$

with $f_{L,E}$ the liquid fraction at the eutectic front, ΔC_E The concentration jump *at the eutectic front*, C_S^E the concentration of the solid eutectic and C_L^E that of the melt in contact. As C_L^E is assumed to be close to the eutectic concentration C_E (= 5.7 wt% Ni), Eq. (1a) can also be written

$$\Delta C_g = f_{L,E}(C_E - C_S^E). \tag{1b}$$

Furthermore, the liquid fraction $f_{L,E}$ at the eutectic front is deduced from the global solute conservation in the solid

$$C_0 = (1 - f_{L,E})C_S^{Al} + f_{L,E}C_S^E = f_{L,E}C_S^E. \tag{2}$$

Finally, by introducing Eq. (2) in Eq. (1b)

$$\Delta C_g = C_0[C_E/C_S^E - 1]. \tag{3}$$

C_S^E can be obtained by combining the local solute balance and the condition of local thermodynamic equilibrium at the solid–liquid interface

$$C_E - C_{SE} = -DG_S/mV, \tag{4a}$$

where the temperature gradient down in the liquid grooves is approximated by that in the solid, namely $G_S = G/2$. Then

$$C_S^E = C_E + DG_S/mV. \tag{4b}$$

By introducing Eq. (4b) in Eq. (3), we obtain an expression for the global measured solute jump ΔC_g in Fig. 1b as a function of experimental data

$$\Delta C_g = C_0\left[\frac{C_E}{C_E + G_S D/mV} - 1\right]. \tag{5}$$

Taking $C_0 = 1.5$ wt%, $C_E = 5.7$ wt%, $G_S(= G/2) = 12$ K/cm, $D \cong 3 \times 10^{-9}$ m^2/s (inferred from the value of 3.7×10^{-9} m^2/s recently obtained in microgravity for an Al–3.2 wt% Ni alloy at $T = 700$ °C [20]), $m = -3.51$ K/wt% Ni, $V = 1.5$ μm/s, Eq. (5) gives $\Delta C_g = 0.20$ wt% Ni which is in fair agreement with the discontinuity at the solid–mushy zone in Fig. 1b. Moreover, the liquid fraction at the eutectic front deduced from Eq. (2), $f_{L,E}(= C_0/C_S^E) = 0.30$ on cross section which corresponds to $1 - (1—f_{L,E})^{1/2} = 0.16$ in longitudinal section, is close to the observed eutectic fraction (see Fig. 2d and b in Section 3.2.1).

Steps iii *and* iv: The tip concentration C_t and the depth ΔH of the mushy zone can be estimated under the assumptions of Flemings' model for stationary cellular solidification [19], which in particular neglects the deformation of the isoconcentration lines in the vicinity of the tips. The concentration at the cell/dendrite tip C_t is higher than C_0 and is given by

$$C_t = C_0 - DG/mV. \tag{6a}$$

By taking $G = 24$ K/cm, Eq. (6a) gives $C_t = 2.9$ wt Ni which is very close to the measured peak in Fig. 1b. Eq. (6a) results in a tip undercooling

$$\Delta T = T_L - T_t = DG/V, \tag{6b}$$

where T_L and T_t are the liquidus and tip temperatures.

Then, a relation follows for the thickness ΔH of the mushy zone

$$\Delta H = \frac{T_t - T_E}{G_M} = \frac{m(C_0 - C_E)}{G_M} - \left(\frac{G}{G_M}\right)D/V, \tag{7}$$

where T_E is the eutectic temperature and G_M the temperature gradient in the mushy zone, whose value is intermediate between G and G_S, but certainly closer to G_S as the liquid fraction in the mush is small. From Fig. 1b, the experimental value ΔH is estimated to 6 mm, which gives $G_M(= m(C_t - C_E)/\Delta H) = 16.5$ K/cm. This value is intermediate between G_S and G as expected.

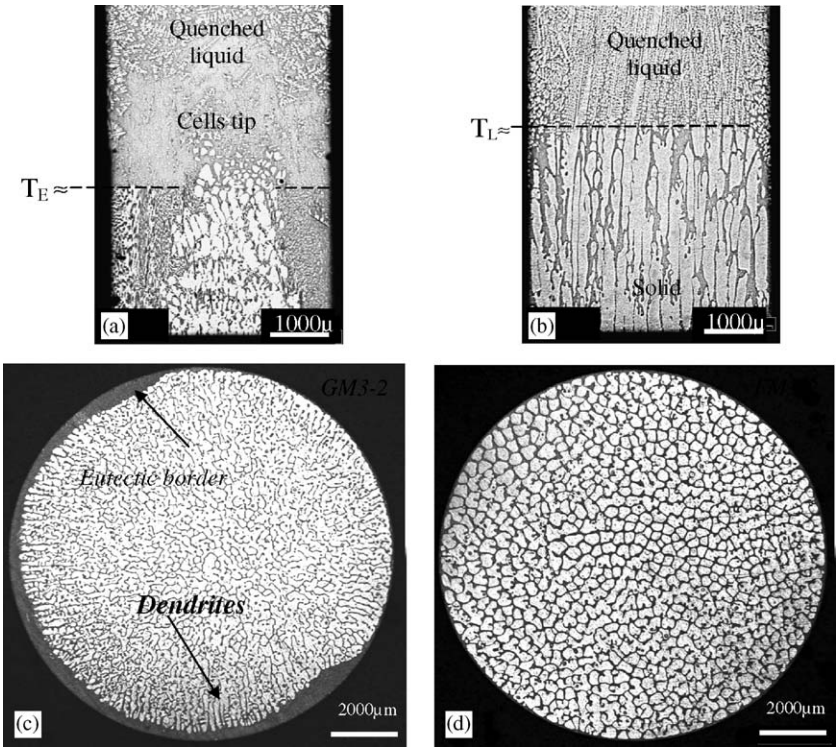

Fig. 2. Comparison of the microstructures obtained in microgravity and in upward solidification on ground for Al–1.5 wt% Ni, $G = 24\,\mathrm{K/cm}$ and $V = 2.3\,\mu\mathrm{m/s}$. (a) and (b) longitudinal sections across the quenched front, (c) and (d) transverse sections below the quenched front. (a,c): GM3-2; (b,d): FM3-2.

The solute concentration profile in the mushy zone C_{MZ} is well described by

$$C_{MZ} = f_L C(d) \tag{8a}$$

with $C(d)$ the solute concentration in the liquid at the location d from the tip ($d \leqslant 0$ for the mushy zone):

$$C(d) = C_t + \frac{G_M}{m} d \tag{8b}$$

and f_L the liquid fraction:

$$f_L = \frac{C_0}{C_0 + (C_S^E - C_0)|d|/\Delta H} = \frac{1}{1 + [(C_E + G_S D/mV)/C_0 - 1]\,|d|/\Delta H}. \tag{8c}$$

Thus, we obtain the variation of the solute concentration in the mushy zone

$$C_{MZ} = \frac{1}{1 + [(C_E + G_S D/mV)/C_0 - 1]|d|/\Delta H} \times (C_t + (G_M/m)d). \tag{9}$$

It is worth noticing that Eq. (9) can be considered as an alternative to Fleming's model, which considers a constant temperature gradient through the whole mushy zone so that it is not compatible with both Eqs. (4) and (6).

Step v: At last, the solute profile in the melt ahead of the tips is classically given by an exponential decay

$$C_{Liq} = C_0 + (C_t - C_0)\exp(-Vd/D). \tag{10}$$

By using Eqs. (9) and (10), an excellent agreement is obtained between the experimental and analytical solute profiles (Fig. 1b).

3.2. Microstructure localisation and radial macrosegregation

3.2.1. Microstructure localisation

Figs. 2a,c and b,d show typical sections for samples solidified at 1 g and in microgravity, respectively. The section through the quenched front in Fig. 2a, flatted parallel to growth, indicates microstructure localisation, namely cell/dendrite clustering towards the centre accompanied by steepling, i.e. the envelope of the tips is strongly distorted and protrudes markedly into the melt ahead of the eutectic front. The detailed shape of this envelope can be reconstructed from successive transverse sections taken across the quenched front (Fig 3a), as illustrated in Fig. 3b which shows its intersections with two perpendicular axial planes. The envelope of the cell/dendrite tips presents a flat central part, curved shoulders, and deeply depressed regions at the periphery. As shown in Fig 2a and in a transverse section taken in the fully solidified part (Fig. 2c), a thin liquid region is present at the periphery down to the eutectic temperature.

Comparatively, no front distortion is observed even at low solidification rates in the microgravity samples: the longitudinal section (Fig. 2b) always shows a macroscopically planar shape of the front envelope, except for the steep but narrow depression at the contact with the crucible which corresponds to the region where heat flows from the crucible into the solid. In transverse section (Fig. 2d), a cellular/dendritic microstructure covers the entire surface and no eutectic border is present.

It results from the ground experiments in the AGHF, complemented with experiments at higher growth rate performed in a laboratory Bridgman furnace with a thermal gradient of 40 K/cm, that the clustering and steepling effects of buoyancy-driven convection progressively decrease as the solidification velocity is increased, so that diffusive-like dendritic growth is achieved beyond a critical pulling rate. Then, there is no longer a peripheral eutectic ring and, alike in Fig. 2d, a large slope of the solid–liquid interface is observed only in a narrow region in contact with the crucible (Fig. 4).

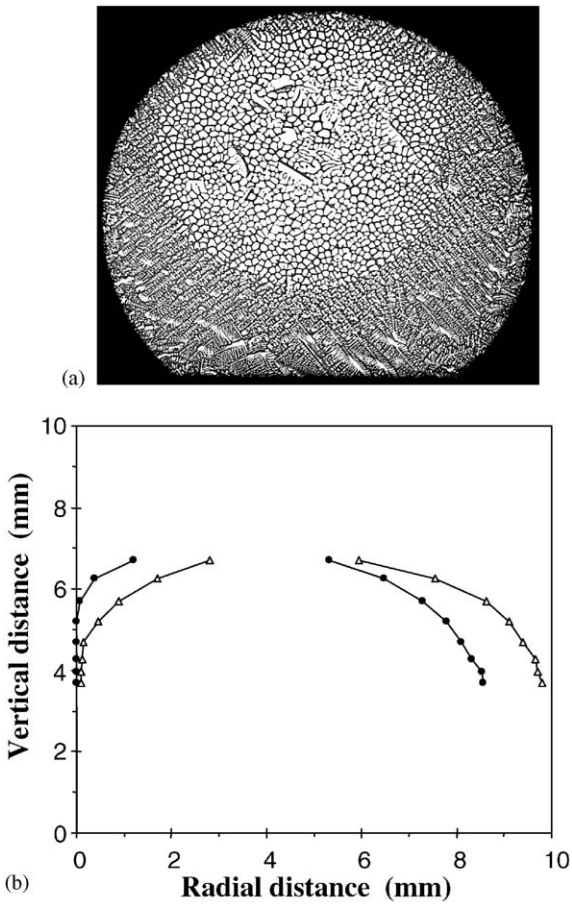

Fig. 3. Front steepling accompanying microstructure localisation in ground samples: (a) example of a transverse section within the quench region (sample GM3-2) and (b) reconstructed front shape (intersections with two perpendicular axial planes). Diameter = 10 mm.

3.2.2. Effect of fluid flow on solute macrosegregation

On the ground, fluid flow results in large radial macrosegregation with negligible longitudinal macrosegregation. This is similar to what was previously found for upwards solidification with a solutal stabilising effect but for the case of a fully dendritic microstructure [21], i.e. a porous mush

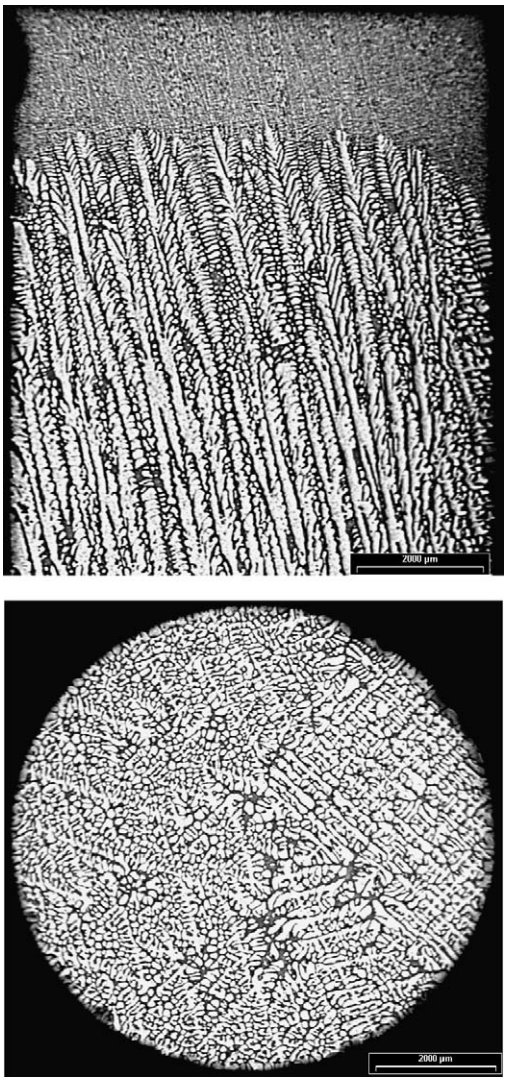

Fig. 4. Longitudinal and transverse sections for a dendritic Al–1.5 wt% Ni sample solidified on ground with $G = 40\,\mathrm{K/cm}$, and $V = 9.5\,\mu\mathrm{m/s}$. Diameter = 8 mm.

and developed sidebranches. In the present experiments, the solute boundary layer adjacent to the tips is swept by the fluid flow driven by the radial temperature gradient due to the higher thermal conductivity of the solid which causes a part of the heat flux transported along the crucible to go into the solid, thus creating a local inward heat flux, a convex shape of the isotherms and a narrow front depression at the crucible wall. This depression is progressively enhanced by the nickel accumulation on the periphery, until the eutectic concentration is reached. A stationary state is eventually achieved because the radial gradient-driven fluid flow is confined in a buffer transported at the front velocity with no mixing with the bulk of the melt, which is the mere reason of the absence of fluid flow effect on longitudinal solute macrosegregation. The most characteristic feature of this state is the clustering of the cellular/dendritic microstructure (e.g. Figs. 2a,c) due to the radial gradient of solute concentration along the solid–liquid interface induced by fluid flow, which leads to competition of the cell/dendrite microstructure with the eutectic pattern.

The radial heat flux into the solid is directly related to the conductivity ratios between the liquid and solid phases, and the crucible. A simple parabolic approximation of the resulting isotherm shape is determined in Ref. [22]. The overall height decrease of a given isotherm while going from the centre to the periphery is then given by

$$\Delta h = R\left[1 - \beta\,\frac{\xi + \gamma}{\beta\xi + \gamma}\right]/2, \tag{11}$$

where R is the sample radius and the controlling non-dimensional parameters are the two conductivity ratios $\beta = \lambda_L/\lambda_S$ and $\gamma = \lambda_C/\lambda_S$ (L, S and C, respectively denoting the liquid and solid alloys, and the crucible), and the ratio of the surfaces of the sample and crucible $\xi = R^2/(e(2R + e))$ with e the thickness of the crucible wall. Taking $\lambda_L = 94\,\mathrm{W/m/K}$, $\lambda_S = 200\,\mathrm{W/m/K}$, $\lambda_C = 20\,\mathrm{W/m/K}$, $R = 5\,\mathrm{mm}$ and $e = 3\,\mathrm{mm}$, Eq. (11) gives $\Delta h = 0.33\,\mathrm{mm}$. This value is more than one order of magnitude smaller than the observed deformation of the front, confirming that, although the radial temperature gradient is responsible for the initiation of the front deformation, it does not control the final amplitude of this deformation which should instead be linked to the solute accumulation just depicted.

The flow pattern is similar to the one previously described [21]. It is worth noticing that, as initially suggested in Ref. [14], the solute enriched, and thus denser, liquid present at the locally depressed front tends to settle in this depression and further retards solidification there. A pocket of stratified

liquid is then formed where the concentration is slightly lower than at the interface at same height. The final depth of the deformation and resulting radial segregation are mainly determined by the amount of solute which can be transported laterally. The effect is greater in the case of a highly permeable dendritic mush where the flow of the strongly enriched interdendritic liquid makes the main contribution to this transport [10]. The present situation is intermediate between that of a porous dendritic mush and that of a planar solidification front. Indeed, it is mainly the flow in the solute boundary layer that, by penetrating into a thin region of the mush beneath the envelope of the cell/dendrite tips, also washes nickel out of the mush. Knowing Δh, estimates can be obtained for the radial temperature gradient, $G_R = G\Delta h/R$, and the Grashoff number which gives the strength of thermal convection [23], $\mathrm{Gr} = \beta_T g G_R R^4/\nu^2$ with β_T $(= 1\times10^{-4}/\mathrm{K})$ the coefficient of thermal expansion of the melt, g the acceleration of gravity and ν $(= 4\times10^{-7}\,\mathrm{m^2/s})$ the kinematic viscosity. For $G = 25\,\mathrm{K/cm}$, $G_R = 1.7\,\mathrm{K/cm}$ and $\mathrm{Gr} = 650$, so that the order of magnitude of the flow velocity in the bulk $\nu\mathrm{Gr}^{1/2}/R = 0.2\,\mathrm{cm/s}$.

4. The solidification microstructure

Microstructures are observed on transverse sections taken in the stationary solidification region for each velocity. When the solidification microstructure was frozen by a quench, the morphology of the solid/liquid region is examined in detail by performing successive transverse sections across the quenched solid–liquid interface (FM3 ad GM3). In the vicinity of the tips, the distribution of primary spacing is measured, and the cellular or dendritic character of the morphology is determined in term of visible sidebranching. The evolution of the microstructure away from the tips down to the eutectic front is also examined. When the solidification front is not frozen by a quench, microstructures are observed on transverse sections taken in the stationary region of solidification. It results from Laue diagrams that the solidified length was long enough for one grain, or a few grains, with a $\langle 1\,0\,0\rangle$-direction within 10° of the vertical pulling direction to impose upon the other grains. This makes our data representative of the $\langle 1\,0\,0\rangle$-orientation, which is most generally considered in modelling and numerical simulations.

4.1. Microstructure in the diffusive regime

For the microgravity samples, primary spacings can be measured on the whole transverse sections, for a large number of cells. Despite the diffusive conditions, the standard deviation of the primary spacing remains large, about 10–20%. This is illustrated in Fig 2d for the case of run FM3-2. The spacing distribution is shown in Fig. 5. A similar result was obtained in Al–Li alloys solidified in microgravity conditions [16]. Thus, the present experiments again confirm that, as theoretically emphasised by Ben Amar and Moussalam [24], there is no selection of a unique wavelength during directional solidification. Yet, the average spacing can be determined together with the minimum and maximum spacings (FM3-1: $\lambda_{min} = 190\,\mu\mathrm{m}$, $\lambda_{max} = 510\,\mu\mathrm{m}$; FM3-2: $\lambda_{min} = 150\,\mu\mathrm{m}$, $\lambda_{max} = 600\,\mu\mathrm{m}$).

In relation with the non-uniform spacing, the cellular/dendritic array is not uniform across the

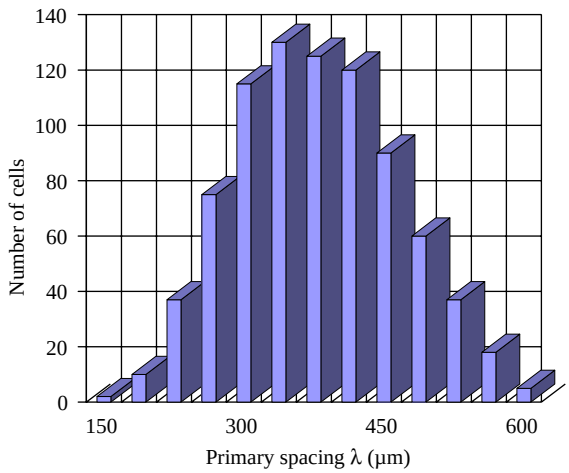

Fig. 5. Distribution of primary spacing for the space experiment FM3-2.

section (Fig 2d). Both smooth cells with a small primary spacing and dendrites with larger spacing exhibiting secondary branches are observed as previously found in Al–Li [16].

These observations confirm the soundness of the selection of experimental conditions around the cell–dendrite transition using the criterion obtained by applying to cells the approach developed for Saffman–Taylor fingers [3,25]. This criterion relates the value of the departure from the level of constitutional supercooling at the cell–dendrite transition v_{C-D} to k, namely

$$v_{C-D} = V_{C-D} m C_0 (1-k)/DGk = [\text{Order } (1) + k]/k \quad (12)$$

with V_{C-D} the critical rate at the transition, which in the limit of small k is equivalent to the transition value proposed from a marginal stability condition by Kurz and Fisher [26]. It follows from Eq. (12) that V_{C-D} does not depend on k in the limit of vanishing k, which is applicable to dilute Al–Ni alloys,

$$V_{C-D} = \text{Order}[DG/(mC_0)]. \quad (13)$$

This criterion compares well with our experiments. Indeed, Eq. (13) predicts V_{C-D} of the order of 1.4 μm/s in our experimental conditions for $G = 24$ K/cm, and the interface microstructure for $V = 1.5$ μm/s (Fig. 2b,d) falls right in the transition, exhibiting both cells and dendrites.

As cells are characterised by the fact that constitutional supercooling at the tip CS_{tip} (relative difference at the tip between the equilibrium temperature gradient, $mC_t(k-1)V/D$, and actual temperature gradient in the liquid, G) is a small quantity [3,25], the criterion (12) was established by arguing that the onset of sidebranching occurs when constitutional supercooling at the tip, i.e. the driving force for the development of corrugations on the cell cap, becomes of order unity, namely

$$CS_{tip} = \{[m\ C_t(k-1)V/D] - G\}/G = \text{Order } (1). \quad (14)$$

It results from the value of C_t in Fig. 1b that this condition is fulfilled as $CS_{tip} = 1.1$, which proves the consistency of the approach.

The Al–Ni benchmark data obtained in space contrast with the reference data obtained by Esaka and Kurz on thin succinonitrile–acetone samples [27], for which the analogy with viscous fingering was never found [25,28], in particular at the cell–dendrite transition because the spacing Peclet number $Pe_\lambda = \lambda V/2D$ is of order unity there. Indeed, it should be noticed that, concomitantly with CS_{tip} of order unity, Pe_λ is much less than one (Table 2) so that the two conditions for the analogy between the directional solidification microstructure and viscous fingers are, for the first time as far as we are aware, simultaneously satisfied. Remembering that $k = 0.0087$, it may be conjectured that the smallness of Pe_λ, through relatively small primary spacings, is advantageous to more efficiently evacuate the large amount of solute rejected upon solidification, from the caps into the liquid groove around. Besides, the cells and dendrites are constrained to adopt a finger shape by the presence of eutectic surrounding their base [29], which is not the case in the absence of eutectic phase where the width of the cells and dendrites, i.e. the solid fraction, has to increase progressively in the liquid grooves to eventually fill the primary spacing in full.

The mean, minimum and maximum primary spacings measured on the whole cross section of microgravity samples at $G = 24$ K/cm, λ_{mean}, λ_{min} and λ_{max}, respectively, are reported in Table 1. These $\langle 1\,0\,0 \rangle$-data are benchmarks in the limit of diffusion transport that are worth comparing with the predictions of the model of Hunt and Lu [4]. Indeed, Hunt and Lu's modelling and numerical simulation closely corresponds to the present experimental situation, in particular the growth direction is $\langle 1\,0\,0 \rangle$ with fourfold cubic anisotropy and neither a selection principle is assumed nor the cell/dendrite shape imposed so that dynamical microstructure selection is fully active. For the sake of clarity, two points should be emphasised, namely (i) sidebranching is not addressed so that all the shapes are smooth, and (ii) the 3D numerical simulation is restricted to axisymmetric shapes, assumed a reasonable approximation for a cell/dendrite in an array. Therefore, it is worth noting that in Hunt and Lu's terminology, cells and dendrites refer to smooth shapes with a

Table 2
Characteristic microstructure parameters for the present experiments on Al–1.5 wt% Ni (FM for experiments in the AGHF Flight Model, GM for experiments in the AGHF Ground Model): mean primary spacing λ_{mean}, minimum and maximum primary spacings λ_{min} and λ_{max}, spacing standard deviation $\Delta\lambda$, spacing Peclet number Pe_{λ}, constitutional supercooling at the tip CS_{tip}, dimensionless width of the solute boundary layer Δ and relative fluid-flow downward velocity $U_{\parallel}/V$.

Sample	Thermal gradient (K/cm)	Pulling rate (μm/s)	λ_{mean} (μm)	λ_{min} (μm)	λ_{max} (μm)	CS_{tip}	Pe_{λ}
FM1-1	34	3.83	331				
FM1-2	34	1.83	—				
FM2	34	0.83	285				
FM3-1	24	2.33	378	190	510	1.7	0.15
FM3-2	24	1.5	394	150	600	1.1	0.10
				$\Delta\lambda$(μm)	Δ	$U_{\parallel}/V$	
GM1-1	35	3.83	242	30	0.53	0.87	0.15
GM1-2	35	1.79	225	35	—	—	0.067
GM2	33	1.0	288	50	1	0	0.048
GM3-1	25	2.67	251	25	0.44	1.27	0.11
GM3-2	25	1.67	205	25	0.27	2.69	0.057

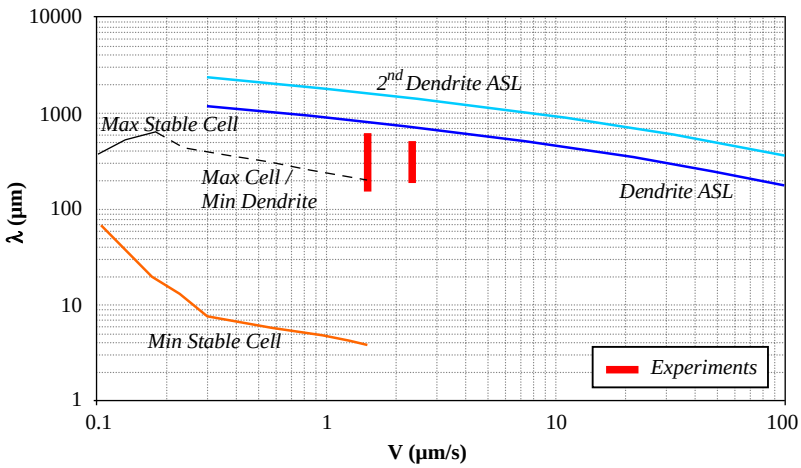

Fig. 6. Plot of primary spacing limits against growth velocity. Numerical calculation with Lu and Hunt model [4] for Al–1.5 wt% Ni, $G = 24\,K/cm$, $\varepsilon_4 = 1.5\%$. Experiment FM3-1: $V = 2.33\,\mu m/s$, $\lambda_{min} = 190\,\mu m$, $\lambda_{max} = 510\,\mu m$; Experiment FM3-2: $V = 1.5\,\mu m/s$, $\lambda_{min} = 150\,\mu m$, $\lambda_{max} = 600\,\mu m$.

rounded tip and smooth shapes with a parabolic tip, respectively. This terminology is used henceforth. The main predictions are the following: (1) above the onset of morphological instability, the range of stable single-cell size and the range of stable single-dendrite size can be computed, (2) these stable single shapes must then satisfy array stability, from which the Array Stability Limit (ASL) is determined (Fig. 6). The ASL gives the minimum stable spacing in an array, i.e. the minimum spacing which does not suffer elimination by larger neighbours because the solute flux at the wall is no longer inwards.

The numerical predictions for Al–1.5 wt% Ni in Fig. 6 are obtained for $G = 24\,K/cm$, with the Gibbs–Thomson coefficient $\Gamma = 1.44 \times 10^{-7}\,K \cdot m$

($\Gamma = V_M \gamma_{SL} T_m / \Delta H_m$, with V_M the molar volume, γ_{SL} the solid–liquid interfacial energy, T_m the melting temperature and ΔH_m the heat of melting, is estimated for pure Al with $\gamma_{SL} = 0.158\,J/m^2$ [30], $T_m = 933\,K$, $\Delta H_m = 10800\,J/mol$ [31], and $V_M = 1.06 \times 10^{-5}\,m^3$) and the surface energy anisotropy parameter $\varepsilon_4 = 1.5\%$. No cell ASL is found because the Maximum Cell/Minimum Dendrite limit occurs earlier (above that limit cells have a tip undercooling larger than dendrites so that they are unlikely). Thus, only $\langle 100 \rangle$ dendritic spacings have a range of stability. Fig. 6 shows that the dendrite spacings at the ASL are about two times larger than the values measured on the microgravity samples. Yet, the predicted tip undercooling, 5.5 K for the growth parameters of FM3-2, is close to that in the experiment $m(C_t–C_0) = 4.9\,K$ for $C_t = 2.9\,wt\%$ Ni.

The result of the sensitivity checking of the numerical predictions to the various alloy parameters in the model is that the variations in the predictions remain small except from unrealistic values, and the experimental primary spacing are never obtained. Therefore, another explanation of the discrepancy should be sought for. A first possibility is an influence of sidebranching. This seems not to be the determinant, as in the experiments, the smooth shapes have smaller primary spacing, and the side arms that form at larger spacings are never pronounced. A second one might be that the experiments do not correspond to the computed dendrite branch, i.e. the experimental shapes are not parabolic or/and not axisymmetric. On the one hand, these two points are difficult to precisely assess because the cuts in the longitudinal sections (e.g. Fig. 2b) go through the very tip only by chance, so that the impression of rather rounded caps given by Fig. 2b might be misleading. On the other hand, it is clear that the panorama of the branches of shapes in Fig. 6 should be completed with the branches of non-axisymmetric, i.e. parity broken, cell/dendrite solutions. Neglecting the fact that the growth direction departs slightly from $\langle 100 \rangle$, to convince the reader it is enough to refer to the results obtained for crystal growth in a channel [32,33]. In particular, besides the complexity of the merging of the branches (see e.g. Fig. 3 in Ref. [32] giving delicate and intricate (Peclet number, undercooling)-diagrams, it results that the bifurcation to parity-broken solutions takes place in the vicinity of the merging point with minimum undercooling. Then, the asymmetric shapes grow at lower undercooling than the symmetric ones, at least for some domain of Peclet number. Furthermore, using the relations (34) and (35) in Ref. [33] to roughly estimate the sensitivity of the spacing to asymmetry for the parameters of FM3-2, suggests that a small amount of asymmetry is enough to increase significantly the spacing. The qualitative transposition to directional solidification is sketched in Fig. 7, with a branch of non-axisymmetric cells.

In the case of Fig. 7, a transition from axisymmetric cells to non-axisymmetric cells occurs rather than the mere transition to dendrites. Then, the non-axisymmetric-cell ASL may be reached before a transition to dendrites, i.e. at a spacing lower than the spacing at the dendrite ASL in Fig. 6.

Although no sound and definitive conclusion can be made without complete quantitative numerical simulation for the experimental conditions, we believe that asymmetry may be the critical missing factor for the following physical argument deduced from our previous 3D-experiments on Pb–30 wt% Tl alloys where we systematically made the

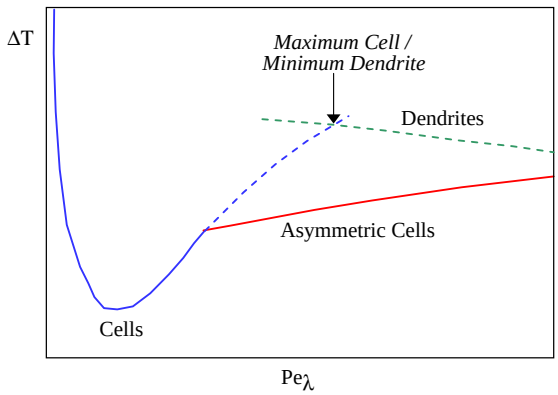

Fig. 7. The cellular and dendritic branches of steady-state solutions including the transposition from Ref. [32] of the lowest branch of asymmetric cells sketched in the diagram giving the tip undercooling ΔT versus the spacing Peclet number Pe_λ. Dashed lines are unstable.

statistical analysis of topological defects and array disorder [17,34]. We found that, during both their dynamical formation from morphological instability and their asymptotic state, the cell/dendrite arrays could be described by a honeycomb on which a large amount of disorder (displacement of cell centres from their ideal positions) is superimposed. Cells/dendrites generally possessed 5–7 first neighbours and should thus be characterised by 5–7 first-neighbour distances (or half-spacings). This suggests that for discussing the ASLs, the solute exchange with all the neighbours has rather to be evaluated, all the more as the cell/dendrite tips have also some freedom to move laterally which could be used to reach a balanced solute exchange with the neighbours. The net result of such a lateral tip drift during array formation would be nothing but non-axisymmetric cells/dendrites.

4.2. Microstructures in convecto-diffusive conditions

In relation with the clustering and steepling phenomena caused by convection on ground as discussed in Section 3.2.1, the ground samples show a radially non-uniform microstructure (Fig. 2a), cellular in the core of the sample and becoming more and more dendritic when approaching the eutectic border at the periphery. Moreover, a strong tendency for these dendrites to grow laterally towards this eutectic region is observed in the region where the local slope of the front of dendrite tips is high (Fig. 2c). This lateral growth is consistent with the fact that the liquid in the depressed liquid region is slightly undercooled due to the horizontal solute gradient.

Due to this radial non-uniformity, it is difficult to perform meaningful measurements of primary spacing. Nevertheless, mean primary spacing was estimated for these experiments in the central cellular region where the fluid flow is locally directed towards the mush. Thus, the average primary spacings were calculated for a limited number of cells (about 40–50) but the standard deviation $\Delta\lambda$ of the measurements was approximately the same as in the case of the microgravity samples. For each velocity, the average primary spacing measured in ground-based experiments is lower than the one measured in space experiments (Table 1), similar to the cellular–dendritic solidification of Al–Li alloys where the convective configuration was solutally destabilising [16]. In the present case, this reduction can be clearly correlated with the inward direction of the flow using the relationship derived by [35]

$$\lambda = \lambda_0(1 + U_\parallel/V)^{-1/2}, \tag{15a}$$

where $U_\parallel$ is the downward component of fluid velocity at the edge δ of the solute boundary layer and λ_0 the primary spacing measured in space under diffusion transport. Eq. (15a) gives

$$U_\parallel/V = (\lambda_0/\lambda)2 - 1. \tag{15b}$$

The weaker reduction of primary spacing compared to fully dendritic solidification [10] can then be attributed to the lower flow penetration in a mush of limited permeability, which is reflected by rather small values of the relative fluid-flow downward velocity $U_\parallel/V$ (Table 2). The component $U_\parallel$ of fluid velocity is related to the dimensionless parameter $\Delta = \delta V/D$ by [36]

$$\Delta = 1/(1 + U_\parallel/V). \tag{16}$$

Eq. (16) means that at δ the diffusive velocity D/δ compensates the overall transport velocity towards the growth front $V + U_\parallel$. The values of Δ are given in Table 2.

5. Conclusion

Comparative experiments in Space and on Earth have been carried out on Al–1.5 wt% Ni alloys, which present a stabilising solutal effect in directional upward solidification on ground. The investigated range of growth parameters was that for which the morphology changes from cellular to dendritic. This comparative experimental study has shown that the coupling between the natural convection and the formation of solidification microstructure can be strong even when solidification is performed in a both thermal and solutal stabilising configuration.

In the reference microstructures solidified under diffusive conditions in microgravity, the standard

deviation of primary spacing is found to remain relatively wide (10–20%). Shapes without and with sidebranches are simultaneously observed in a given cross section, the later being associated with larger spacings. This mixed interface microstructure is found in a range of control parameters for which the level of constitutional supercooling at the tips in the liquid is of order one. The comparison with the numerical predictions of Lu and Hunt's model reveals that the experimental primary spacing is about half of the theoretical value. This misfit led us to consider a few possible shortcomings that might provide an explanation. It is suggested that the full panorama of the branches of symmetric and non-symmetric cellular and dendritic shapes, together with their array stability, is the prerequisite for sound and definitive discussion. We think that this is more especially important in the complex and delicate regions where there are branches close by or merging.

By comparison, ground samples show a strong macroscopic deformation of the front and associated radial non-uniformity of the microstructure, i.e. microstructure localisation. This effect is in between those observed during similar directional solidification experiments performed respectively with fully dendritic microstructures, and below the morphological instability threshold with a smooth solid–liquid interface. It is due to a fluid-flow pattern driven by the radial temperature gradient, causing macroscopic deformation enhanced by solute accumulation. In relation with this radial solute segregation, the microstructures appear cellular in the core and dendritic at the periphery. In addition, the primary spacings measured in the central part are smaller than in the corresponding microgravity experiments, which can be related to the fact that the flow velocity has a component oriented towards the mush in this region.

Acknowledgments

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